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The role of plants in ironstone evolution: iron and aluminium cycling in the rhizosphere

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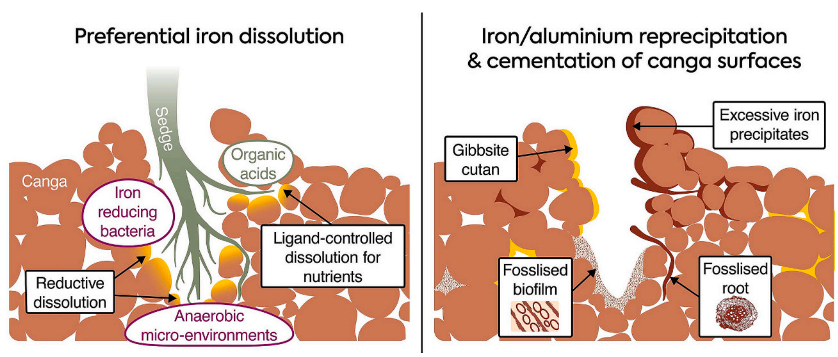
HIGHLIGHTS

- High magnification characterisation of canga rhizosphere.
- Alternate Fe and Al precipitation caused by plant-induced geochemical changes.
- Plant roots contribute to iron oxide precipitation *via* three different processes.
- Plant root exudates preferentially dissolve iron, enriching aluminium.
- Plant-induced Fe cycling contributes to the evolution and stabilisation of canga.

GRAPHICAL ABSTRACT



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ABSTRACT

The Carajás plateaus in Brazil host endemic epilithic vegetation ("campo rupestre") on top of ironstone duricrusts, known as canga. This capping rock is primarily composed of iron(III) oxide minerals and forms a physically resistant horizon. Field observations reveal an intimate interaction between canga's surface and two native sedges (*Rhynchospora barbata* and *Bulbostylis cangae*). These observations suggest that certain plants contribute to the biogeochemical cycling of iron. Iron dissolution features at the root-rock interface were characterised using synchrotron-based techniques, Raman spectroscopy and scanning electron microscopy. These microscale characterisations indicate that iron is preferentially leached in the rhizosphere, enriching the comparatively insoluble aluminium around root channels. Oxalic acid and other exudates were detected in active root channels, signifying ligand-controlled iron oxide dissolution, likely driven by the plants' requirements for goethite-associated nutrients such as phosphorus. The excess iron not uptaken by the plant can reprecipitate in and around roots, line root channels and cement detrital fragments in the soil crust at the base of the plants. The reprecipitation of iron is significant as it provides a continuously forming cement, which makes canga horizons a 'self-healing' cover and contributes to them being the world's most stable continuously exposed land surfaces. Aluminium hydroxide precipitates ("gibbsite cutans") were also detected, coating some of the root cavities, often in alternating layers with goethite. This alternating pattern may correspond with oscillating oxygen

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concentrations in the rhizosphere. Microbial lineages known to contain iron-reducing bacteria were identified in the sedge rhizospheric microbiome and likely contribute to the reductive dissolution of iron(III) oxides within canga. Drying or percolation of oxygenated water to these anaerobic niches have led to iron mineralisation of biofilms, detected in many root channels. This study sheds light on plants' direct and indirect involvement in canga evolution, with possible implications for revegetation and surface restoration of iron mine sites.

1. Introduction

Ironstone, ferruginous duricrusts, canga, ferricrete and iron laterite are the common names used to describe the capping rock that covers many of the world's supergene iron ore deposits (Ramanaidou and Morris, 2010; Salles et al., 2019). This goethite-cemented breccia is physically and chemically resistant to weathering, armouring the underlying friable saprolite (Dorr, 1964; Monteiro et al., 2018b). The protective rock blanket has resulted in topographically elevated plateaus. These outcrops support unique ecosystems with high endemism (Carmo and Jacobi, 2016; Silveira et al., 2016), currently undergoing massive habitat loss due to intensive opencast mining (Salles et al., 2019). There is evidence to suggest that microbiological processes contribute to the formation of this capping rock (Monteiro et al., 2014; Gagen et al., 2019a; Levett et al., 2020b), yet the role of the epilithic flora in canga evolution have not been evaluated.

In this study, we focus on canga from the tropical southeast Amazonian Craton in Carajás, Brazil. This consolidated duricrust is composed of banded iron formation (BIF) clasts and iron pisoliths, cemented by secondary goethite and gibbsite (Monteiro et al., 2018a; Spier et al., 2019), resulting in iron concentrations of >45 % and aluminium concentrations >3 % (Gagen et al., 2019a). Canga is relatively porous; however, the presence of small seasonal ponds indicates it has low permeability (Dorr, 1964). Due to its variable composition and high aluminium concentrations, canga is often treated as a by-product of the mining process (Dorr, 1964; Silva et al., 2018) and blended with high-grade ore (Spier et al., 2019).

Canga can be found exposed at the surface or covered by Mesozoic or Cenozoic rubble (Monteiro et al., 2018a). Pockets of thicker soil cover are common in depressions where organic matter accumulates (Nunes et al., 2015). These canga soils are nutrient-poor, relatively acidic (pH <5) and offer poor water retention and mechanical support for plants (Messias et al., 2013; Fernandes, 2016). Hence, the 'mountaintop grassland' is comprised of open herbaceous-shrubs and hemicryptophytes (Jacobi and Carmo, 2011; Gibson et al., 2015), that are juxtaposed by the surrounding rainforests (Skirycz et al., 2014; Silveira et al., 2016). The harsh edaphic conditions on canga control the rupestrian plant communities, enhancing plant diversity and habitat specialisation (Poot et al., 2012; Messias et al., 2013).

Cangas from the Carajás plateaus are some of the longest continuously exposed formations on earth (Monteiro et al., 2014). The evolution of canga includes (i) slow *in situ* hydration of hematite; (ii) downward leaching of soluble species; (iii) iron metasomatism of weathered material (*i.e.*, canga soils); and (iv) reoccurring dissolution-reprecipitation of iron (Beauvais, 2009; Monteiro et al., 2018a). Geochronology of goethite cement in canga revealed that the pattern of dissolution and reprecipitation has continued intermittently for the last ~80 Ma (Monteiro et al., 2018a). Since the solubilities of iron(III) oxide minerals are very low at circumneutral pH, it is assumed that most of the dissolution and transportation of iron in canga requires reduction to the ferrous [iron(II)] state followed by precipitation of insoluble ferric oxides (Ramanaidou and Morris, 2010).

Although some iron dissolution can occur abiotically (Dorr, 1964), significant reductive dissolution at the near-surface environment it likely the result of biogeochemical processes (Parker et al., 2013; Monteiro et al., 2014; Gagen et al., 2019a). In anoxic conditions, iron-reducing bacteria can use iron(III) as a terminal electron acceptor for microbial respiration, resulting in significant quantities of reduced, soluble iron(II) (Kostka and Nealson, 1995; Weber et al., 2006). The aqueous iron(II) can subsequently enhance, by electron transfer, the reductive dissolution of iron(III) oxide minerals from within the mineral lattice (Handler et al., 2009). Changes in oxidation potential or the occurrence of iron-oxidising bacteria at the redox boundary can result in the recrystallisation of hydrous iron oxides. Regardless of their metabolism, bacterial cells can also serve as nucleation sites for iron oxide formation (Mera et al., 1992; Li et al., 2013). The presence of fossilised biofilms (Levett et al., 2020a; Levett et al., 2020b) corroborate the role of microorganisms in canga formation.

Dissolution and reprecipitation of iron minerals can also be a rhizospheric process resulting from the release of root exudates (*e.g.*, simple organic acids, flavonoids and phytosiderophores) and the plant-driven oxygen and pH fluctuations (Colombo et al., 2014; Terzano et al., 2015). These exudates, together with protons secreted from the roots and decaying organic compounds, create conditions that favour mineral weathering by way of ligand-promoted and/or acid-reductive dissolution of iron oxide minerals (Hinsinger et al., 2003; Terzano et al., 2015). In iron-rich conditions such as canga, non-mycorrhizal plants may acidify the rhizosphere and excrete large amounts of exudates, in order to acquire other nutrients, such as phosphorus, that are commonly associated with goethite (Neumann and Römheld, 1999). Plants may also facilitate the percolation of solutions containing oxygen, inducing iron oxidation, resulting in formation of iron-lined root channels (Verboom and Pate, 2006) or iron plaque coatings (Fresno et al., 2016).

Previous restoration attempts of disrupted canga flora have had limited success (Fernandes, 2016; Silveira et al., 2016), likely due to the loss of the physical characteristics of the duricrust (Salles et al., 2019). While revegetation studies often focus on plant survival with topsoil salvaged from the mined area, they fail to consider that many native canga plants grow directly on the canga surface or in rock crevices (Carmo and Jacobi, 2016). Furthermore, plants affect the physical properties of the canga on which they grow, and may also assist with remediation attempts, especially regarding surface stabilisation (Gagen et al., 2019a; Paz et al., 2020).

Despite multiple studies hypothesising about the contribution of the flora to iron cycling in canga (Pate et al., 2001; Monteiro et al., 2014; Gagen et al., 2019a), the mechanism and extent of plant involvement in canga formation remains unclear. This understudied aspect is surprising, considering that field observations of plants growing on canga suggest an intimate involvement of roots and their secretions in canga evolution (Monteiro et al., 2018a; Gagen et al., 2019a). Therefore, targeting iron dissolution and reprecipitation processes, we investigated the plant-rock interaction by examining roots and root channels of native plants growing on canga and the soil crust at the base of these plants. We analysed this system using a multidisciplinary approach, including electron microscopy, X-ray absorption and infrared microspectroscopy to characterise the iron mineralogy. In addition, 16S rRNA gene amplicon sequencing was used to identify indigenous rhizospheric microorganisms.

2. Methodology

2.1. Field site and sample collection

Hand specimens of canga were collected at Serra Norte, in the Carajás mineral province, Pará, northern Brazil, S06°00′50.2″ W050°17′53.8″ (Fig. 1A). Sampling concentrated on exposed surfaces

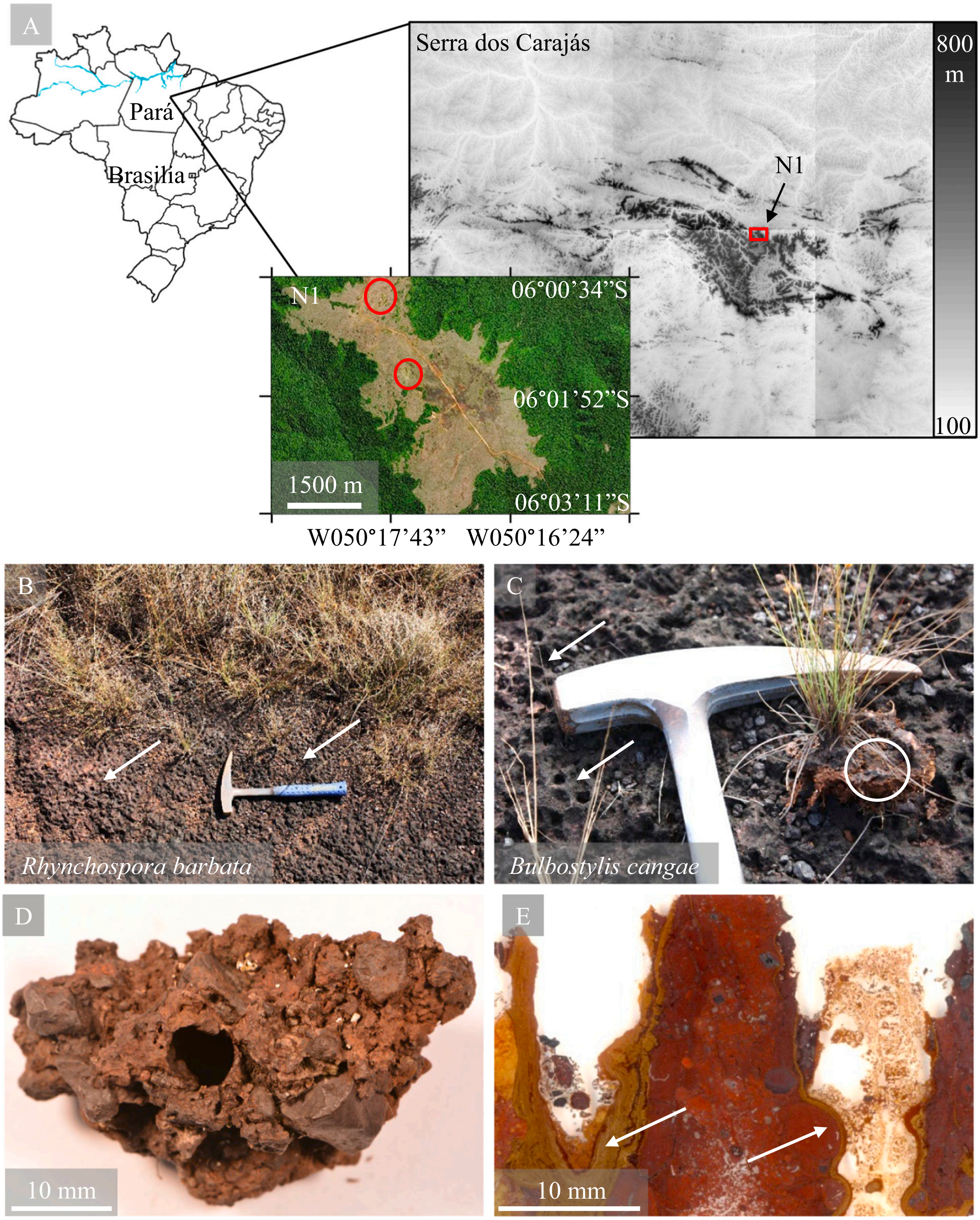

Fig. 1. Geographical context of ironstone outcrops in Carajás plateaus, Brazil and field observations of plants growing in this rock, locally known as canga. Topographic map (US Geological Survey, digital elevation using SRTM, https://earthexplorer.usgs.gov) with satellite images showing areas of sample collection (A, circles). Small indentation texture was commonly observed (B-C, arrows) around the base of two endemic sedge species (*Rhynchospora barbata* and *Bulbostylis cangae*, B and C respectively) suggesting a double role of plants in the dissolution and reprecipitation of canga. The white circle in panel C marks the ‘skirt’ at the base of the sedge, where the images in Fig. 4C and Fig. S5 were taken. Underneath such partially organic ‘skirts’, individual clasts are strongly cemented together by secondary iron oxides, seen here in a hand sample (D); this hard, consolidated crust had to be broken out using a hammer and chisel. Layers of secondary goethite can be seen in the vertical cross-section of the indentation texture (E), lining the walls of the root-channels, and cementing exogenous fine-grained materials that had washed into the cavity (E, arrows).

associated with roots of native plants on the N1 plateau. Most plants in this ecosystem do not grow directly in the canga but instead grow in cracks or in small islands of accumulated pebbles and organic matter (*i.e.*, canga soil) (Carmo and Jacobi, 2016). In this study, we focused solely on plants that grow directly through canga rock (*i.e.*, rupestrian). This was operationally determined by uprooting plants and discarding those found to be utilising existing crevices or where the roots were associated with soil pockets (*e.g.*, plants from the *Bromeliaceae*, *Convolvulaceae* and *Fabaceae* and families). This process resulted in the selection of only two plants from the family *Cyperaceae* (sedges), namely *Rhynchospora barbata* and the recently described *Bulbostylis cangae* (Nunes et al., 2017).

While *Rhynchospora barbata* is widespread in Central America, West Indies and the northern parts of South America (Strong, 2006), *Bulbostylis cangae* is endemic to the canga of the Carajás (Nunes et al., 2017). However, both plants occupy a similar niche at the edge of shallow perched lakes or at poorly drained areas inundated following heavy rain events. The canga rock hosting these sedges contained sharp indentation texture (*e.g.*, Fig. 1B-C), with depression diameter of 1–2 cm and depth of 3–10 cm (*e.g.*, Fig. 1D-E). We collected 10 samples from the base of each sedge species and 10 samples from the surrounding canga currently unoccupied by plants.

Canga samples collected in the vicinity of the sedges *Rhynchospora barbata* and *Bulbostylis cangae* possessed a macroscopic organic indentation texture (Fig. 1B-C, arrows). While most of these depressions were hollow, uprooting existing plants revealed that some of the depressions were active root channels. At the base of the sedges, a 'skirt' of organic-rich crust is often observed (see circled area, Fig. 1C). This crust, composed of detrital particles cemented by organic matter and iron oxides, is competent and hard, requiring a hammer to break it from the underlying canga (Fig. 1C-D). A vertical cross-section of the indentation texture revealed that the walls and bottom of these root channels are lined with secondary layers of goethite, frequently cementing fine exogenous particles (Fig. 1E).

2.2. Chemical analyses

Bulk rock samples were analysed for elemental composition and mineralogy. For this, samples were crushed and milled to a fine powder (<63 μm) using an agate ball mill. Major elemental concentrations, determined by inductively coupled plasma atomic emission spectrometry (ICP-AES) after acid digestion (Advanced Laboratory Services, Stafford, Australia), were: 42.8 % iron, 8.2 % aluminium, 0.5 % titanium and 0.06 % phosphorus. Note, that canga composition varies greatly, even between samples collected in close proximity from each other (Gagen et al., 2019a). Plant-available micronutrients, measured after diethylenetriamine penta-acetic acid (DTPA) extraction (according to Rayment and Lyons (2011)), were (mg/kg): 132.8 iron, 2.06 manganese, 0.52 copper and 0.42 zinc. The mineral assemblage, analysed by X-ray diffraction (XRD, Bruker D8 Advance), was: hematite, aluminium substituted goethite and gibbsite. The pH, measured in a 1:5 soil:water suspension with deionised water after agitation for 3 h, ranged from 4.5 to 5.4. For this purpose, canga soil is defined as ~1 g friable material in or surrounding the roots channels. Further laboratory quality checks are described in Supporting Information.

2.3. Electron microscopy

Rock samples from the base of sedges and from the surface canga with indentation texture (*i.e.*, active and inactive root channels) were impregnated with epoxy resin (EMbed 812), without any dehydration steps. Samples containing organic components (*i.e.*, roots and soil crust) were fixed using 2.5 % glutaraldehyde (final concentration), dehydrated through an acetone dehydration series, and embedded using an epoxy infiltration series. The epoxy-embedded samples were made into petrographic cross-sections (100 μm thick), polished to a 0.5 μm finish and coated with 15 nm carbon deposition. To increase the contrast of the roots against the epoxy background, one of the root-containing cross-sections (used in Fig. 2D) was vapour-stained using 4 % osmium tetroxide solution. Characterisation of the microstructure of consolidated canga was performed using a JEOL-7100 field emission scanning electron microscope (FE–SEM). Backscatter electron (BSE) micrographs of the root-rock interface of living sedges or empty root channels were taken at an accelerating voltage of 15 keV.

2.4. Elemental and mineralogical mapping

Cross-sections imaged by SEM were subsequently mapped using the X-ray fluorescence microscopy (XFM) beamline at the Australian Synchrotron (Clayton, Australia), equipped with a 384-element Maia detector in backscatter geometry (Ryan et al., 2018). The cross-section was scanned through an 18.5 keV monochromatic X-ray beam focused to a 2 μm × 2 μm spot using a Kirkpatrick-Baez mirror pair with 2 μm step size in both the horizontal and vertical directions and a 0.33 msec per pixel effective dwell time with a total photon flux of approximately 1.6×10^9 photon/s. Scans were conducted on-the-fly in horizontal direction with discrete steps in the vertical direction. The resulting elemental maps (~10 mm × 5 mm) were analysed using GeoPIXE (CSIRO, 2015) and quantified using manganese, iron and platinum standard foils.

To examine the iron speciation at the root-rock interface, laterally resolved iron K-edge X-ray absorption near-edge structure (XANES) data was collected at the XFM beamline from regions of interest (~500 μm × 200 μm) selected form the XFM maps. Mapping elemental speciation using 'XANES imaging' differs from the normal approach of μ-XANES as it potentially allows XANES spectra to be extracted from every pixel in the map (Etschmann et al., 2014). The XANES data were collected with the same scan parameters as the XFM maps using 148 non-uniformly spaced energies across the iron K-edge from 6.962 to 7.462 keV. Data were preconditioned to remove pixels with poor signal using the data collected at an incident energy of 7.462 keV. The remaining per-pixel spectra were normalised to an edge-step of one, before a set of two pseudo-Voigt function were fitted to the rising edge of each spectra (Berry et al., 2010). Spectra with a poor fit ($R^2 < 0.98$) were omitted from further consideration. The energy where the normalised intensity of the rising edge reaches a value of 0.8 is linearly dependent on the degree of iron oxidation, with the entire range from reduced to oxidised encompassing 6 eV (Berry et al., 2018; Jones et al., 2020). In the present case, we use the energy where the fit crosses 0.8 of normalised intensity to determine the degree of iron oxidation to reduce the influence on noise. Representative spectra from high and low iron(II) regions are shown in Fig. S1. Data are presented with a false-colour table ranging most reduced to most oxidised, with the energy span of the represented data included to indicate the range of values.

Since the lateral distribution analysis of lighter elements (*e.g.*, aluminium) requires a vacuum, cross-sections were also mapped by micro X-ray fluorescence (μXRF) using an IXRF ATLAS X, configured with a 50 W XOS Flexbeam source with a molybdenum anode and two KETEK H150 detectors (total photon flux of approximately 2×10^8 photon/s). Scans were conducted at 50 kV/1000 μA tube power, with a 25 μm resolution. Each pixel value is the sum counts on both detectors across that pixel for the energy region corresponding to the major peak of aluminium. Further quality checks for the elemental mapping are described in Supporting Information.

The mineralogy of new iron oxide precipitates was determined using Raman spectroscopy, as described in Levett et al. (2020c). In short, Raman spectra were collected using WITec Alpha 300 Raman microscope equipped with a 532 nm laser and operating at ~0.5 mW. The Raman maps, representing the relative concentration of the detected minerals, were produced with 1 μm spot size and 16 s dwell time per spectrum. Following background removal, the Classical Least Squares (CLS) method was used to construct the maps. Each spectrum was deconvoluted into its mineral constituents by using reference spectra of hematite, goethite, lepidocrocite and anatase. Data analysis was

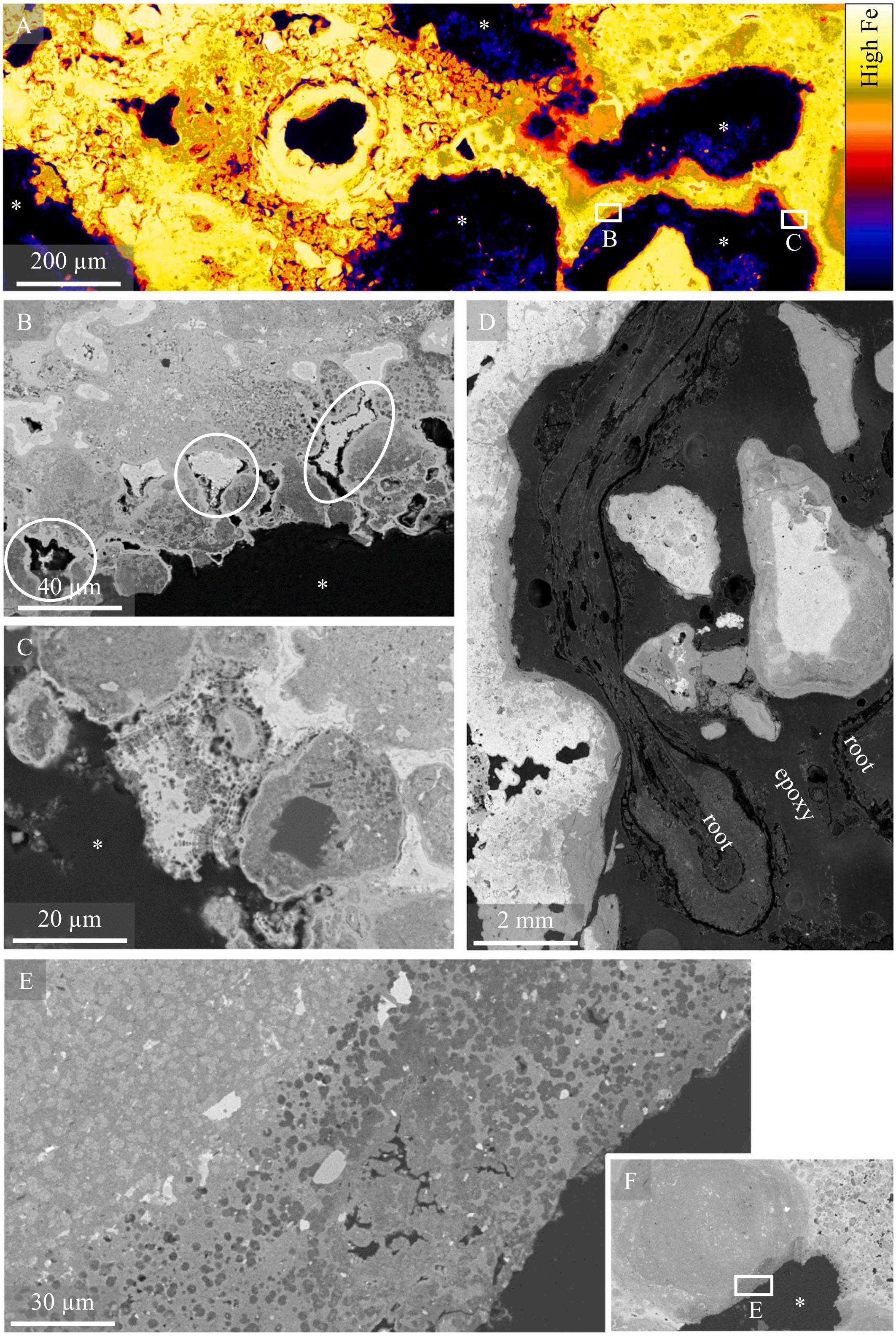

Fig. 2. Cross-sections of root channels, possessing living plants at the time of collection, demonstrating mineral weathering. The depletion of iron at the edge of the cavities was evident in the XFM map of iron distribution (A, red-orange) and in the backscatter electron micrographs of inserts (B) and (C). Circles in (B) highlights the dissolution of iron-rich areas. A root burrowing through canga (D) had contributed to the aluminium enrichment rim at the root-rock interface. A weathered pisolitic grain (*E*-F) showing preferential iron leaching features with an atomic mass gradient from aluminium-rich area at the cavity edge toward brighter, iron-rich areas further from the cavity. In all backscattered electron micrographs, black is the carbon-based embedding epoxy or root, dark grey is aluminium-rich and bright white is iron-rich. Asterisks (*) indicate active root channels.

conducted using WITec Project Four software.

2.5. Microbial community analysis

The rhizospheric prokaryotic communities for each of the sedges were characterised, with the intention of identifying the organisms that contribute to iron cycling in the root channel environment. For this, 50 ml of water was injected into root channels (occupied by the sedges) and after 10 s, 20 ml of this solution was collected *via* syringe and passed through a Sterivex filter. This procedure was repeated for three root channels using the same filter, resulting in one composite sample for each of the sedges. Next, 2 ml of Lifeguard soil preservation solution (Qiagen) was added to filters before transporting back to the laboratory for later DNA extraction. DNA was extracted from the cut filters using the DNEasy Powersoil kit (Qiagen), yielding ~5 ng/μl. The V6-V8 region of the 16S rRNA gene was amplified using universal primers 926F and 1392r and sequenced on a MiSeq Sequencing System (Illumina). Sequences (23,010–27,465 per sample) were processed using MOTHUR1.43 (Schloss et al., 2009) according to a modified MiSeQ SOP (Gagen et al., 2018) and using the Silva reference database SSU Ref NR 99 v132 (Quast et al., 2013; Glöckner et al., 2017). Species diversity was estimated as Shannon's diversity index. Sequences have been submitted to GenBank Sequence Read Archive under BioProject PRJNA598517.

2.6. Organic acids

Flushing the root channels with water also allowed for the detection of major organic exudates. As above, 50 ml of water was injected into currently occupied root channels (not the same root channels used for DNA collection), followed by the immediate collection of 20 ml of filtered (<0.22 μm) solution into sterile sealed vials. The procedure was repeated for three root channels for both plant species. The samples were kept frozen (−20 °C) until they were concentrated by solid phase extraction (SPE) and analysed by ion chromatography mass spectrometry (IC-MS) at the Chemistry Centre Laboratory, Queensland Government Department of Environment and Sciences (Gokhale and Rohrer, 2003). The limit of quantification (LOQ) was 0.025 mg/l.

3. Results

Microscale characterisation using multidisciplinary, non-destructive techniques allows for the mapping of the same regions to gain an in-depth understanding of the mechanistic processes that contribute to iron cycling in the canga rhizosphere. The characterisation results are interlinked and corroborative; as such, they are presented as inorganic mineral processes (Sections 3.1–3.2) and organic biotic processes (Sections 3.3–3.4).

3.1. Iron oxide dissolution

Dissolution of canga in the rhizosphere does not progress as a uniform front. The iron map of the root channels obtained using XFM (Fig. 2A) highlights the iron concentration at the root-rock interface is lower than in the bulk rock, with concentrations gradually increasing farther from the root (Fig. S2). Iron dissolution at the edge of the root cavity is corroborated by the BSE-SEM micrographs (Fig. 2B-C), where iron-rich areas (shown as brighter colours) are the first to be dissolved (Fig. 2B). As a result of iron leaching, roots burrowing through canga leave behind an aluminium enriched region around the root tunnel (Fig. 2D). This relative depletion of iron and enrichment of aluminium was confirmed by EDS analysis (Fig. S3). The sequence of canga dissolution is best demonstrated in Fig. 2E-F, where the centre of the grain is mottled with iron-rich speckles and the preferential mobilisation of iron from the weathered edge has resulted in a dappled texture enriched in aluminium. The circumference of the grain is not intact, with some of the grain fully dissolved by plant-induced weathering processes.

The distribution of iron(II), extrapolated from the synchrotron-based XANES imaging, reveals that the root-rock interface is depleted in total iron (Fig. 3B) as well as iron(II) (Fig. 3A). Here the ~3 eV change in the XANES map of the rock allows the possibility that iron is not present in either fully reduced or oxidised state. However, focusing on the organic components of the system (Fig. 3C-D), the full oxidation range is observed, where the most reduced iron is coating the roots and the soil organic matter.

3.2. Reprecipitation of iron and aluminium oxides in canga

In the canga rhizosphere, mobile iron may reprecipitate between the roots' outer layers (Fig. 4A-B), completely coat the root's surface (cast fossilisation, Fig. 4C), or completely infill and preserve the internal structure of the roots (permineralisation, Fig. 4D-E). Iron oxide minerals follow or infills the root structures (*e.g.*, Fig. 4A-B), indicating of iron has precipitated out of solution.

Inactive root channels in canga commonly contain fossilised microbial biofilms coating the iron-depleted edges (Fig. 5A-D). Bacteriomorphic structures were determined to represent remnant microorganisms based on cell size (1–2 μm), rod shapes and evidence for cell replication (Fig. 5D and G, arrows). Microbial fossilisation in canga is well documented and can be associated with titanium mobilisation (Levett et al., 2021). Often, a gradient in the degree of iron mineralisation of the biofilms is observed, where the most exposed bacterial layer is the least mineralised, reflective of the fact that it is a younger or growing surface. Raman spectroscopy indicated the bacterial mineralisation is hematite with microcrystalline anatase inclusions (Fig. 5C). The thickness of the mineralised biofilms can range between 5 and 70 μm (data not shown). Chemical precipitation of iron oxide minerals is also commonly observed in inactive root channels, sometimes growing on top of weathered fossilised biofilm (Fig. 5E-G). These acicular precipitates are lepidocrocite (Raman spectroscopy, Fig. 5F); they nucleate on the edge of the root cavity and grow inward to fill the pore space, with typical thickness of ~100 μm. Synchrotron-based XFM analyses of trace metals (Fig. S4) revealed diverging elemental behaviour around fossilised microbial biofilms, with: (i) vanadium, a bioessential element, enriched in the hematitic biological precipitates; (ii) mild chromium and cobalt enriched in the chemical lepidocrocitic precipitates; and (iii) titanium and manganese depleted from all analysed secondary precipitates.

Secondary aluminium precipitates were also detected in many of the active root channels (Fig. 6). These precipitates, confirmed by Raman spectroscopy to be gibbsite, have smooth continuous texture and they follow the morphology of the cavity circumferences (Fig. 6A-C). Frequently, bands of gibbsite and iron hydroxide occur in an alternating pattern (Fig. 6D-G); different numbers of alternating of aluminium/iron hydroxide generations were detected within the 3 cm^3 hand sample. The secondary iron precipitates appear to act as a cement for the fine detrital grains (Fig. 6E).

3.3. Rhizospheric microbial communities

In many of the active root channels examined, electron microscopy revealed that recent bacterial mats were freshly mineralised, with a density gradient of less mineralisation closer to the rhizosphere (*e.g.*, Fig. 7A-C). The most frequently detected bacterial 16S rRNA gene sequences in these rhizospheric environments were from the *Acidobacteria*, *Proteobacteria* and *Planctomycetes* phyla, followed by sequences not classifiable below the domain level, *Chloroflexi* and *Actinobacteria*. (Fig. 7D), reflecting typical soil environments. Total bacterial and archaeal sequences were clustered into 1687 and 1686 unique OTUs (at <0.03 % distance) for *Bulbostylis* and *Rhynchospora* rhizospheres, respectively. The 40 most abundant OTUs (Fig. 8) comprised only ~30 % of the total number of unique OTUs in each sample. Indeed, the high Shannon diversity index of 6.3–6.5, indicates very diverse microbial

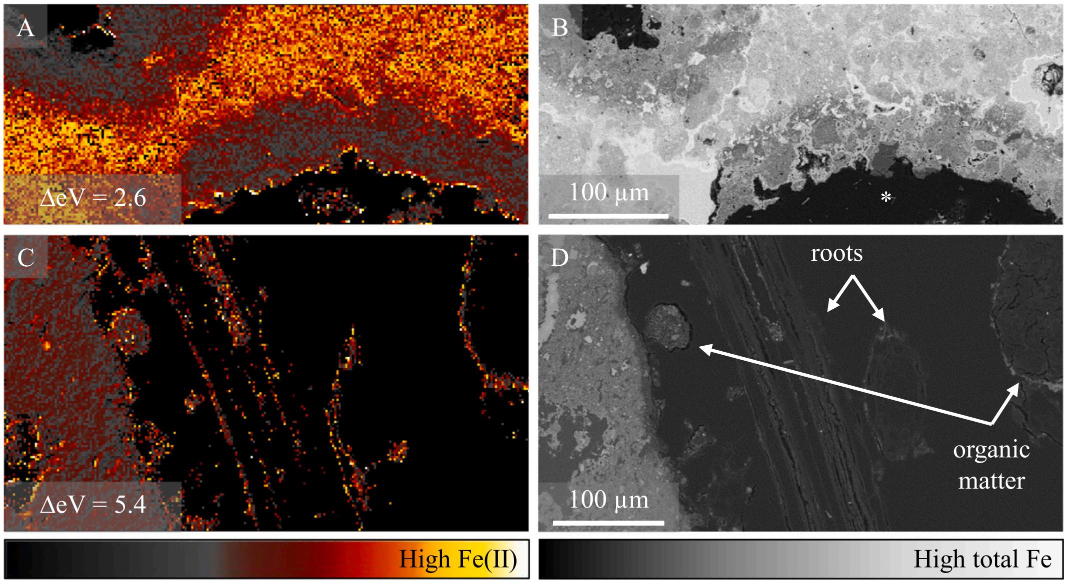

Fig. 3. Lateral distribution of reduced iron as measured by XANES, with matching backscatter electron micrographs for comparison. Note the full oxidation range of iron is detected when the difference in the energy of the absorption edge is ~6 eV. At the root cavity edge, iron depletion corresponds with minimal iron(II) concentration (A-B). However, iron(II) is observed complexed or sorbed to the root surface and organic matter (C-D). Black areas lack iron; asterisks (*) indicate active root channels.

communities with many low abundance members. The majority of OTUs were from lineages without close cultured relatives (<95 % identity to named isolates in the public domain; Fig. 8). Most OTUs that demonstrated >97 % rRNA gene identities were of widespread, slow growing, chemoorganotrophic soil bacteria (*e.g.*, *Koribacter versatilis*). Two OTUs (9 and 19; Fig. 8) showed identity (though only 95.6 % and 96.8 % across the region sequenced) to known acidophilic iron-reducing bacteria, *Acidicaldus sp*. MK6 and *Acidibacter ferrireducens*, respectively (Kozubal et al., 2012; Falagán and Johnson, 2014). The known iron-reducing lineage *Anaeromyxobacter* (OTU 82) was also detected in both environments and comprised 0.1 % and 0.3 % of the total sequences in each library, respectively. Sequences from known iron-oxidising lineages were not abundant in either of the rhizospheric communities.

3.4. Organic acids

Of the measured organic acids, oxalic acid had the highest concentrations in both sedge rhizospheres ($p < 0.001$), followed by acetic, lactic, and formic acids (Fig. 7E). Citric, maleic and shikimic acids were not detected. The exudate profile was similar between the plants ($p > 0.5$).

4. Discussion

4.1. Plant-associated weathering drives iron oxide precipitation

Most plants that grow on canga do not grow directly in the rock but creep along the surface, utilise gaps in the rock or grow in pockets of accumulated soil and organic matter (Salles et al., 2019). However, we observed two native sedges species (*Rhynchospora barbata* and the newly described endemic *Bulbostylis cangae*) that grow directly in canga by burrowing into the host rock. These species compose the majority of the canga shrubland in inundated fields and in the periphery of shallow perched lakes. This environmental niche is characterised by low coverage of trees, large patches of exposed rock surface and seasonal flooding or surface runoff.

Previous observations of concentric goethite bands in tubular cavities and ferruginised organic matter have been proposed to represent plant-affiliated circulation of iron in canga (Monteiro et al., 2014). Here, we provide clear evidence of direct root-induced dissolution and associated reprecipitation of iron and aluminium oxide minerals in the canga rhizosphere. The modern surface feature of the small indentation texture and the prevalence of goethite-lined root channels (Fig. 1) demonstrate this dual role plants play in iron cycling in canga. Alongside the known role of plants in iron-bearing mineral weathering (Robin et al., 2008), microscopic characterisation of the root channels highlights the heterogeneous nature of the biological and chemical precipitates (*e.g.*, Fig. 5). These data suggest that these processes are spatially and temporally separated, reflective of frequent (and sometimes rapid) oscillation in local chemical conditions.

Dissolution textures at the root-rock interface extend from the macroscale (*i.e.*, root channels in Fig. 1B-E) to the microscale (*i.e.*, localised etching in Fig. 2B-C and E-F). These dissolution textures have yet to be observed in canga that was not associated with plants. Moreover, this extensive dissolution is difficult to explain by abiotic processes alone, corroborating the root-induced nature of this dissolution texture. Plants can promote mineral dissolution by acidification of the rhizosphere and exudation of organic molecules as chelating agents (Robin et al., 2008; Colombo et al., 2014). While the pH of the root channel is likely lower than our *ex situ* pH measurements of ~5, it is unlikely that root-induced protonation could result in substantial acid-dissolution (Hinsinger, 2001). However, the *in situ* rapid extraction method for analysing organic acids allowed for a semi-quantitative profile of the exudates, in which relatively high concentrations of oxalic acids and other biogenic chelating ligands were detected (Fig. 7). Low molecular weight carboxylates can adsorb reversibly to the metal oxide surface, weakening the metal-oxygen bonds, resulting in the detachment of a metal-ligand complex (Hinsinger, 2001). It is accepted that this ligand-controlled dissolution can result in substantial iron and aluminium dissolution (Cornell and Schwertmann, 2003).

In oligotrophic substrates such as canga, where phosphorus availability is low due to the strong bonding of phosphate by iron oxides, plants need to adapt to acquire this limiting nutrient. The copious exudation of organic acids is a plausible phosphorus acquisition strategy since these ligands can solubilise phosphorus through mineral dissolution of aluminium/iron oxides (Lambers et al., 2003). Moreover, the presence of oxalic acid can decrease phosphorus adsorption on goethite

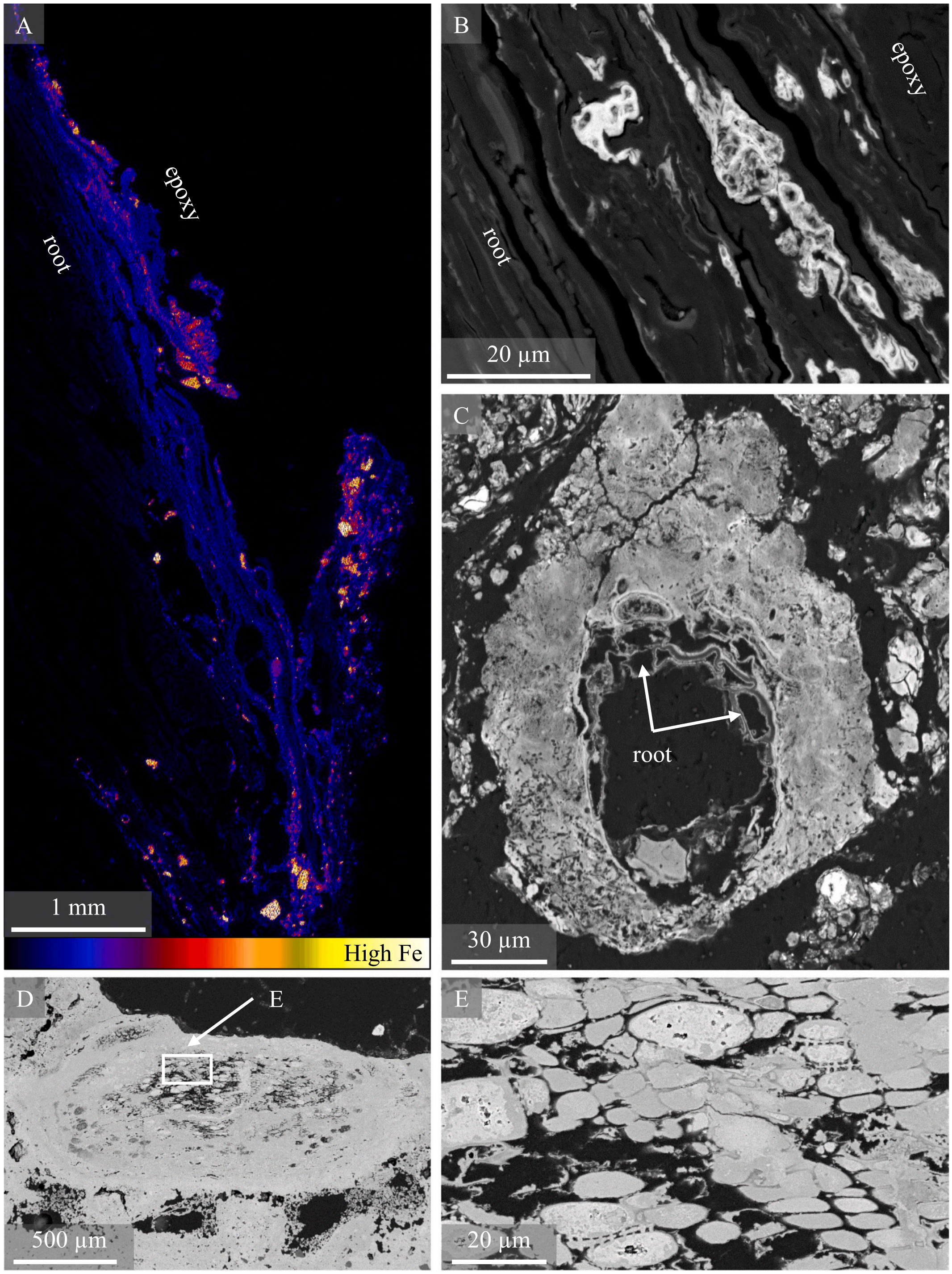

Fig. 4. Mobilisation and reprecipitation of iron in roots at different stages of mineralisation. In the sedges' rhizosphere, dissolved iron from the surrounding canga has started precipitating between the root's outer layers (iron concentration XFM map and backscatter electron micrograph, A and B respectively). In the crust around the sedge, some roots have been partially mineralised, with iron coating the exodermis cell membrane (C). Fully fossilised roots can be found in the top 5 cm of canga, where iron completely infilled the cortex cells (D-E).

and gibbsite (Robin et al., 2008) and is known to correlate with phosphorus deficient growth conditions (Neumann and Römheld, 1999). Additionally, since canga is an aluminium-rich environment with pH values that can be as low as pH ~4.0 (Skirycz et al., 2014), the pronounced exudate pattern could also assist with aluminium complexation and toxicity reduction (Jones, 1998; Chen et al., 2017).

In this study, the concentrations of root exudates other than organic acids (*e.g.*, flavonoids and coumarins) were not evaluated. However, these molecules, common in rhizospheric environments, can act both as ligands and as reductants, promoting reductive dissolution (Perron and Brumaghim, 2009; Mladěnka et al., 2010). Furthermore, it is likely that in the bottoms of the small root channels, where organic matter from previous decomposed plants accumulates, reducing conditions can develop. Reductive dissolution, the occurrence and the fate of the produced iron(II) are discussed in Section 4.3.

In addition to the role of plants in the dissolution of iron oxides, the data presented here reveal plant-associated iron oxide reprecipitation and cementation of canga surfaces. This is caused by the abundant

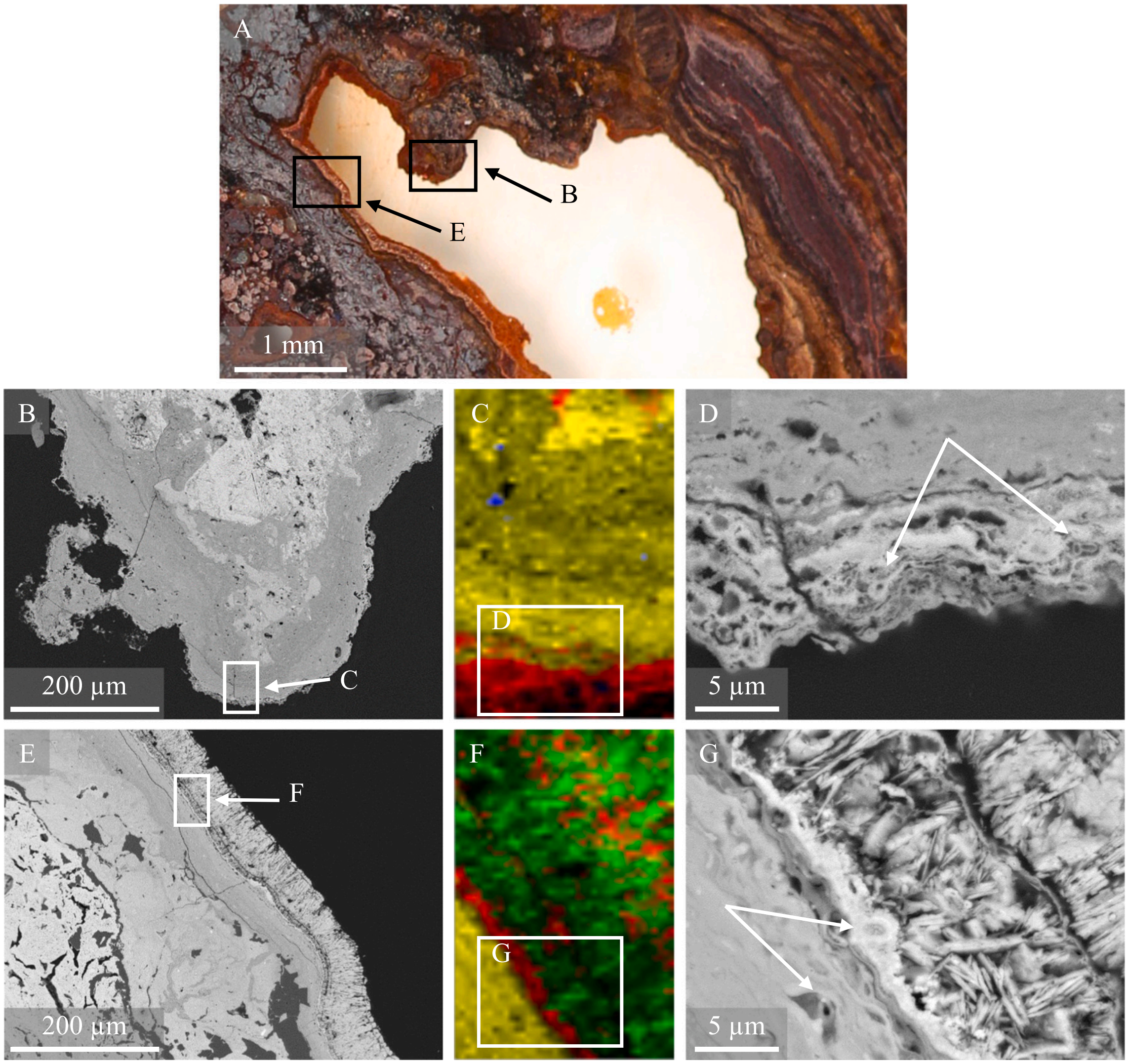

Fig. 5. Biological and chemical precipitation of iron oxides in the root cavity. The complexity of the dissolution and reprecipitation processes can be seen in an inactive root channel (A). Backscattered electron micrographs reveals both mineralised bacteria (B, arrows in D) and chemical, acicular precipitates (E, G) occur within the same cavity. Note how the chemical precipitates have coated a weathered, fossilised biofilm (arrows in G), likely the result of a change of the chemical conditions over time. Raman spectroscopy of the iron precipitates highlight that the biological precipitate is hematite, coating a goethite grain with anatase disseminated throughout the samples (C) and that the chemical precipitate is lepidocrocite (F). Raman colours: yellow = goethite, red = hematite, blue = anatase, green = lepidocrocite.

dissolved iron, excess to plant needs, that reprecipitates as new secondary minerals. Microbial consumption of the organic components of the metal-ligand complexes would result in the release of ionic iron (and aluminium) into the pore solution, and the subsequent redeposition iron (and aluminium) oxides (*e.g.*, Fig. 5 and Fig. 6). The rhizosphere bacterial community analysis (Fig. 8) illustrates the presence of diverse heterotrophic organisms, capable of utilising such organic anions and decomposing the chelated complexes. These observations highlight the effects of root exudates on ferricrete cycling in the rhizosphere of *Cyperaceae* (sedges), previously reported only for *Proteaceae*-dominated communities in Western Australia (Verboom and Pate, 2003).

In canga, fresh iron precipitates can develop on existing features in the substrate and/or coat roots, cracks, and cavities in the root vicinity. The constant plant-driven chemical changes are responsible for the formation of micro-environments within the canga rhizosphere, resulting in a diverse assemblage of iron oxide minerals. The occurrence of the metastable mineral lepidocrocite (Fig. 5F), common in circumneutral environments of alternating reducing/oxidising conditions, implies a temporary formation of iron(II) (Cornell and Schwertmann, 2003).

Titanium was not detected in any of the new precipitates but observed mostly as discrete anatase grains in the goethite matrix, likely inherited from the protolithic units (Monteiro et al., 2018a). In comparison, the chromium and cobalt are slightly enriched within the fresh lepidocrocite precipitates, demonstrating the mobility and re-distribution of these elements within circumneutral environments. The co-localisation of vanadium with the fossilised biofilm supports the suggested role of vanadium as a potential biosignature (Marshall et al., 2017).

Two plant-driven features appear to contribute to iron reprecipitation within the canga rhizospheres. The first is the existence of roots in different stages of fossilisation (Fig. 4), where iron mineralisation of the roots can be attributed to (i) the nucleation of soluble iron on negatively charged root surface (*e.g.*, following the consumption of the organic moiety of the iron-ligand complexes by rhizosphere microorganisms); (ii) the oxidation of iron(II) ions once pH-E_h conditions changes, followed by the precipitation of iron hydroxide at their existent location; and (iii) the passive ferruginisation of dead plant tissue prior to its decomposition, resulting in preservation of the root internal structure. It

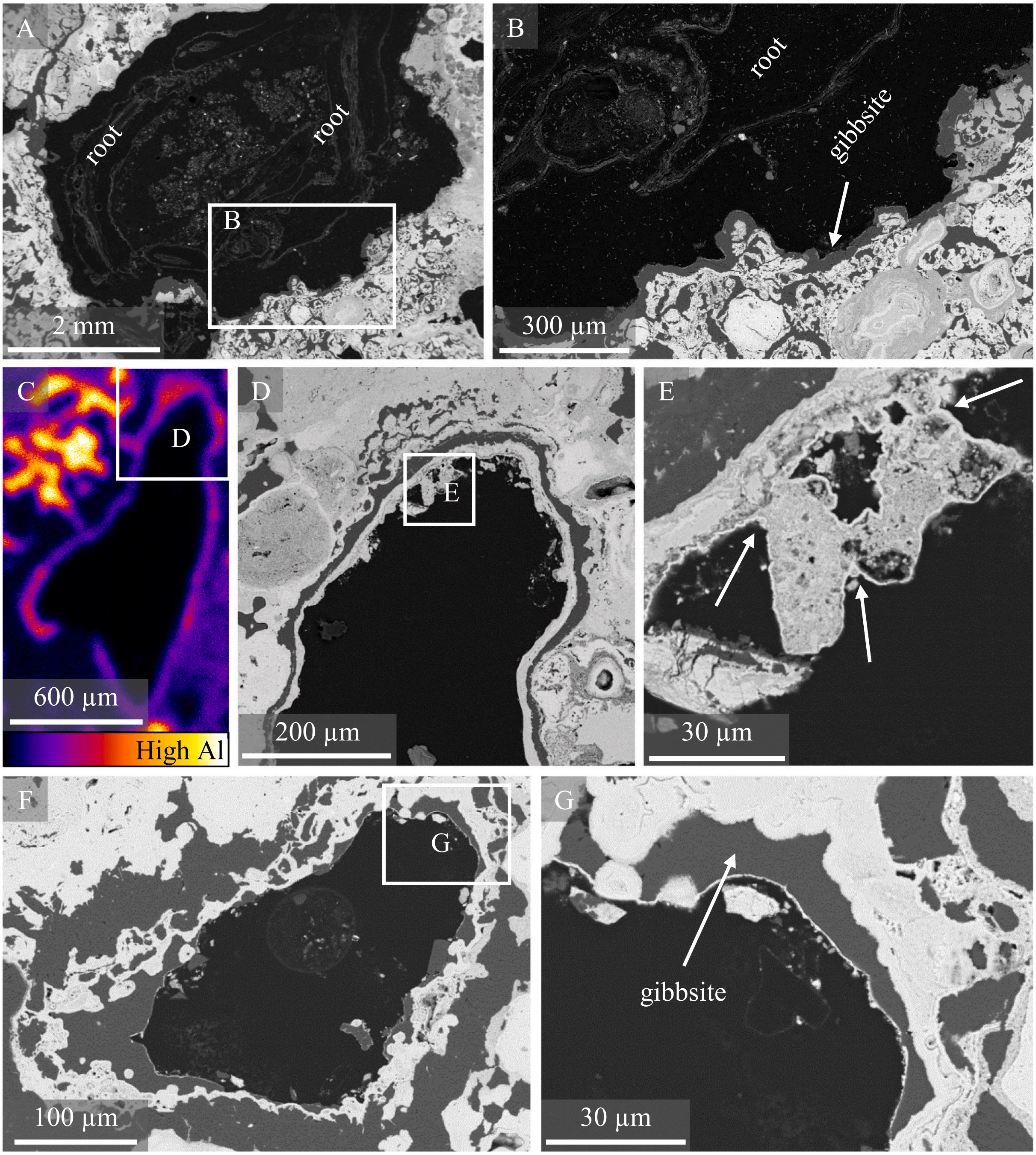

Fig. 6. Secondary aluminium precipitates as gibbsite coatings on the edge of the root cavities. Around live burrowing roots, the gibbsite precipitation is the most recent event, blanketing existing features (A-B). Micro-XRF map showing aluminium is distributed in the cavity circumferences and as pore infilling (C). Often fluctuating conditions results in several alternating bands of aluminium and iron hydroxides precipitates (D-E), sometimes with as many as 5 generations in the same cavity (F-G). Note how iron precipitates act as cement for fine fragments (E, arrows). In the backscattered electron micrographs of canga, gibbsite has a uniform dark grey colour, with smooth texture (*e.g.*, arrows in B and G).

is probable that all three processes have occurred in our surface samples, due to the chemical changes in the rhizosphere micro-environments, resulting in various fossilisation phases.

The second iron reprecipitation feature has resulted in the cementation of the biological soil crust at the base of the sedges (Fig. S5). While part of the cementing agent in the crust is clearly organic, it is also iron-rich. We propose that iron precipitates coat the organic components of the crust as scaffolding. Over time, rhizospheric microorganisms degrade the organic components, yet the iron precipitates retain the structure and anchor the inorganic particles. Recurrence of this cycle may result in an increase of canga thickness and strength, unless the erosion rate accelerates. In this regard, the soil crust and the hard consolidated canga rock can be assumed to represent two end members of the iron deposition that governs canga evolution.

4.2. *Differential behaviour of iron and aluminium oxides in canga*

While many studies have explored the role of iron cycling in canga evolution (Gagen et al., 2018; Anand et al., 2019; Gagen et al., 2019a; Levett et al., 2020b), the contribution of aluminium had been somewhat neglected, doubtlessly due to the higher concentration of iron in the rock. However, a lot can be learnt regarding the geochemical conditions

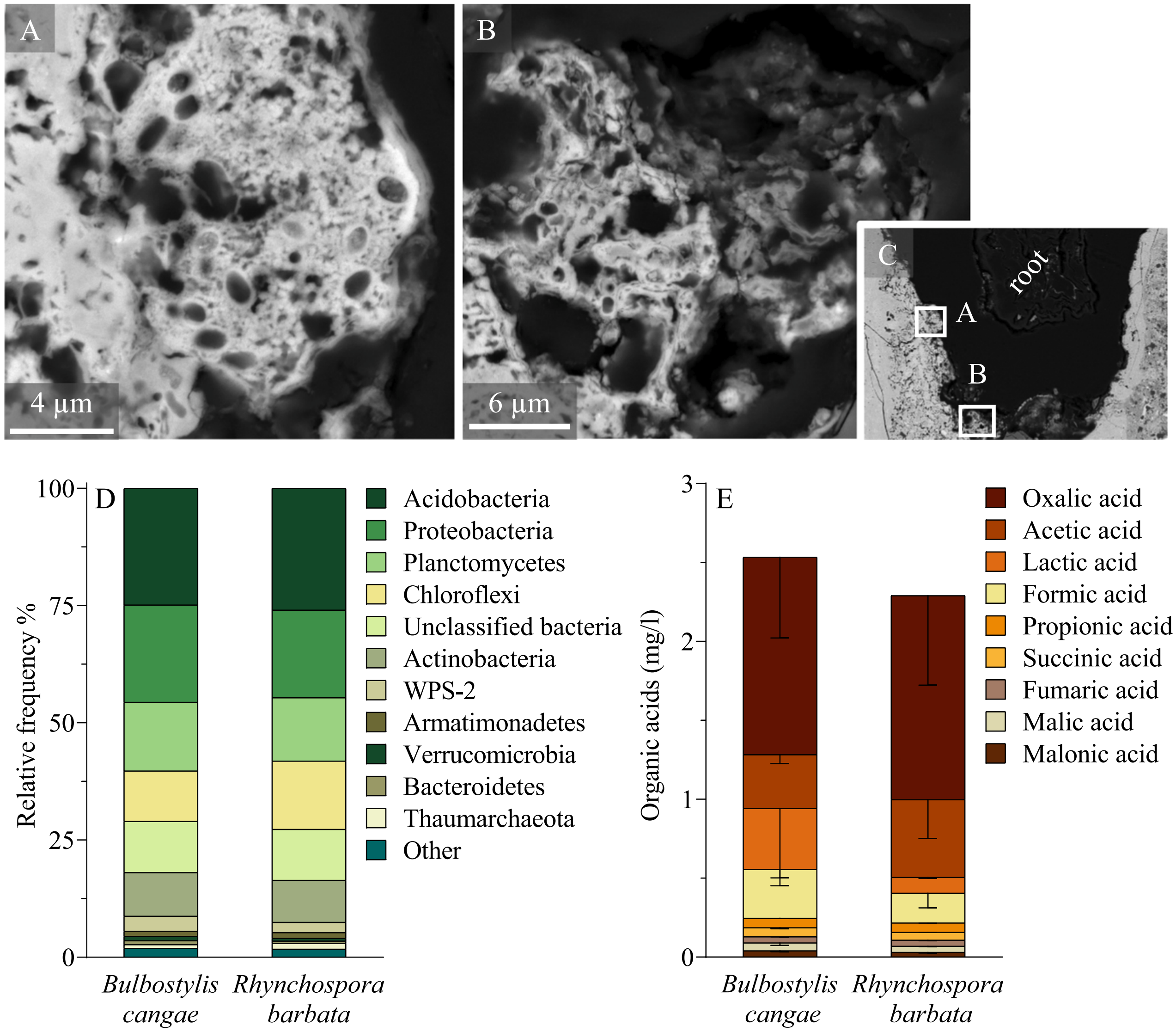

Fig. 7. Evidence of freshly mineralised bacterial mats in the active root channels, with corresponding microbial and exudate profiles. Backscattered electron micrographs demonstrating a mineralisation gradient at the root-rock interface (A-C). Unique 16S rRNA gene sequences, collected from the sedge rhizospheres, are shown, grouped at the phylum level (D). Concentrations of the detected organic acids (E) are presented as means ± SD, *n = 3*.

of the rhizosphere by the different behaviour pattern of these two elements. Goethite in canga has complex textures and composition, with high aluminium substitution contents, especially close to the surface (Monteiro et al., 2018a). Gibbsite is also present (Monteiro et al., 2014; Spier et al., 2019) but without a mechanism to explain its localisation. Here, we investigate two observed processes of aluminium enrichment in the rhizosphere: that of residual aluminium-substituted goethite depleted in iron, and that of gibbsite precipitation.

In most of the root channels examined in this study, iron was preferentially leached at the root-rock interface (Fig. 2); often a clear gradient of iron concentration was visible, ending with an aluminium enriched rim closer to the root. What could be the mechanism behind this preferred iron dissolution? To the extent of our knowledge, separate hydrolysis constants and solubility products for iron and aluminium in aluminium substituted goethite have not been reported; however, due to higher reactivity of aluminium, it is unlikely that iron dissolution is simply a matter of thermodynamic favourability. Association of the phosphorus with the iron-rich areas is also not a probable driver for iron dissolution because phosphorus was shown to colocalises with aluminium in biogenic goethite (Levett et al., 2019). Since most chelation ligands complex with iron as well as with aluminium (Pate et al., 2001), this lack of discrimination cannot explain the preferred iron dissolution either. Likewise, this selectivity in elemental mobilisation is not controlled by pH, since both metals are relatively immobile in circumneutral pH and require similar acidification before acid-dissolution commence. Thus, reductive dissolution is presumably the main driver for the observed refinement in the rhizosphere (*i.e.*, redoximorphic feature), as only iron (and not aluminium) is susceptible to reduction (Vepraskas et al., 2018). This reduction could be promoted by the root secretion of reductants (*e.g.*, flavonoids), iron-reducing bacteria, and/or seasonal saturation of root channels, as discussed in Section 4.3.

The second process of aluminium enrichment in the rhizosphere of canga results in the precipitation of gibbsite cutans (Fig. 6). The weathering of residual kaolinite, aluminium substituted goethite, and gibbsite in canga have released minor concentrations of aluminium ions to the pore water solution. This relatively immobile element has subsequently redeposited as secondary gibbsite, coating existing features and infilling voids and interconnected cracks. For almost pure aluminium hydroxides to precipitate, the large amounts of dissolved iron had to be removed from the pore water solution. This is further evidence of the role of reductants and reducing conditions at the root-rock interface, allowing the dissolved iron to be transported in

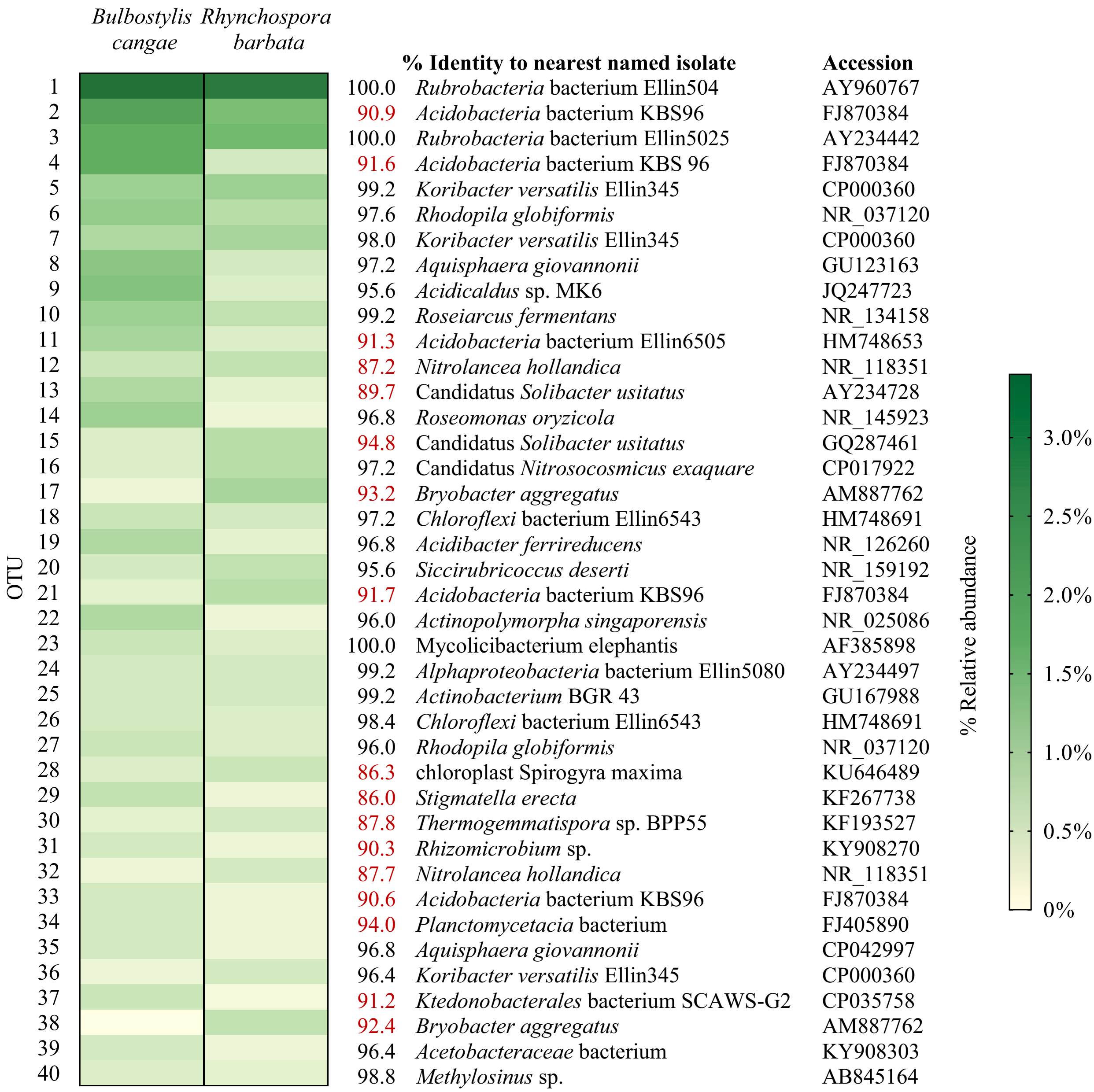

Fig. 8. Heatmap analysis showing the most abundant OTUs in the sedge rhizospheres. The nearest named isolate in the public domain and its accession number (preference given to published studies) are shown for each OTU. Highlighted in red are deep-branched OTUs, with no close cultured representative in the public domain (<95 % 16S rRNA gene identity).

solution toward the lower profile or out of the system, as suggested before by Monteiro et al. (2014). Moreover, leaching of soluble iron and recrystallisation of secondary gibbsite could be considered part of the lateralisation and maturation processes of duricrusts (Nouazi Momo et al., 2019). To this end, aluminium can be pictured as another cementing agent in the rhizosphere, sealing cavities and pores in the iron breccia, strengthening the weathered rock, and contributing to its competency. Although this phenomenon has been demonstrated before for duricrust horizons in soils (McKeague and Protz, 1980), and more recently for subsurface canga (Levett et al., 2020c), to the best of our knowledge, this is the first time that gibbsite deposits are revealed to be directly associated with rhizospheric processes.

Alternating bands of aluminium and iron hydroxides around roots in canga (Fig. 6, D-G) are another example of the oscillating conditions in the rhizosphere, specifically those relating to the reductive status of the micro-environment. The sequence of mineral precipitation implies that soluble iron may reprecipitate when exudation of organic acids and flavonoids stops, leading to deposition of goethitic layer on top of the gibbsite-coated surface. At this stage, cementation of exogenous material (*e.g.*, fine particles and detrital fragments) by iron oxides could occur (Fig. 6E). It is plausible that the imprint of iron coating, including the cemented material, represents a flooding event in the cavity. The sequential coatings of gibbsite over iron oxide layers may indicate that gibbsite precipitation occurred during the recent stages of cavity formation, associated with current burrowing roots and the return of reductive conditions. This differential precipitation pattern could be assigned to late pedogenic processes involving a change in the groundwater reducing conditions or water table level variations.

4.3. Occurrence and fate iron(II) in canga

Iron speciation maps illustrate that iron(II) is preferentially leached from the root-rock interface (Fig. 3A). These data extends recently reported finding that iron(II) is enriched within canga relict grains compared with the surrounding goethitic cements (Levett et al., 2021). The depletion of the labile iron(II) from the root-induced weathered edge supports the hypothesis that the iron(II) in the grains is remnant from the original magnetite minerals (Levett et al., 2021). However, the presence of a thin layer of relatively reduced iron(II) at the very edge of the cavity indicates the dynamic nature of the system, where dissolved iron(II) tends to re-adsorb to exposed surfaces. Interestingly, the organic components of the root-rock interface were coated with mostly reduced iron (Fig. 3B), suggesting that the organic moieties can stabilise iron(II) even in aerobic conditions; for example, after air-drying of the root channels. This could be attributed to the ability of most organic acids and soil organic matter to bind both iron(III) and iron(II) in stable complexes (Suzuki et al., 1992). Most importantly, it underlines the ubiquity of iron(II) in the canga rhizosphere.

The development of reducing conditions in the root channels is presumably a major factor controlling the dissolution of canga. It was technically impractical to measure the concentration of iron(II) of the pore water solution in the rhizosphere of canga; our closest estimations of 1–15 ppm iron(II) were measured in small (~15 cm diameter, ~20 cm depth) ephemeral water bodies in the vicinity of the collected sedges (Gagen et al., 2019a; Levett et al., 2020b). It stands to reason that some of the processes that occur in the small water bodies occur also in saturated root channels, resulting in similar concentrations of iron(II). Once iron(II) has been released into the pore water solution it may act as an important reductant in the presence of organic ligands (*e.g.*, carboxylic acids) driving a synergistic reductive dissolution of iron oxides (Cornell and Schwertmann, 2003; Kang et al., 2019).

Plant exudates could also stimulate microbial metabolisms capable of mobilising iron from the rock. The axiom that the rhizosphere has a higher microbial density and metabolic activity compared to bulk soil is doubly true in canga where soil cover is scarce. The frequent observations of mineralised bacteria of different morphologies and multiple generations at the root channel edges (*e.g.*, Fig. 5B-D and Fig. 7A-C) emphasise the role of plants and their secretions in creating microenvironments suitable for microbial colonisation. Similarly to microbial surveys of canga soils (Vieira et al., 2018; Gastauer et al., 2019), we found the sedges' rhizospheric community to be very diverse, with many slow-growing soil bacteria, commonly correlated with low nutrient and carbon input and low pH (*e.g.*, *Acidobacteria*, *Chloroflexi* and *Actinobacteria*). With regards to microbial iron-cycling, *Acidicaldus* and *Acidibacter*, lineages containing members known to participate in dissimilatory iron reduction (Kozubal et al., 2012; Falagán and Johnson, 2014), were identified in both our samples. The activity of iron-reducing bacteria would require anoxic or microaerophilic conditions, that could conceivably be generated by an association with aerobic heterotrophs in thick biofilms (Gagen et al., 2018). The rhizospheric microbial communities can promote reducing conditions by using plant exudates as electron donors, causing the depletion of dissolved oxygen in the wet root channels. Thus, organic carbon is hypothesised to be the limiting factor in microbial-induced iron reduction in canga (Gagen et al., 2018; Gagen et al., 2019b; Levett et al., 2020b); the addition of organic acids to the system by plant exudation is likely to stimulate indigenous microbial metal reduction in laterites (Newsome et al., 2020). Although low in abundance, a small but consistent population of iron-reducing bacteria can support high concentrations of soluble iron (Levett et al., 2020b), possibly due to the relatively poor crystallinity of the iron oxides in canga (Parker et al., 2013).

Oxidation is inseparable from reduction of iron at circumneutral pH under natural conditions. The absence of known iron-oxidising bacteria in our samples suggests that iron oxidation is presumably a passive process in canga rhizosphere. Hence, elevated concentration of oxygen in the root channels could be seen as a liability for microaerophilic lineages lacking a mechanism to avoid encrustation, leading not only to cell death but their preservations as fossils. We suggest that iron(II)-rich solution at the bottom of the root channel have reacted with surface oxygenated water or atmospheric air, resulting in amorphous iron hydroxide precipitation upon the cellular remains, and subsequent transformation to stable secondary minerals (*e.g.*, hematite and goethite). Alternatively, electrostatic adsorption of iron cations on the negatively charged cell envelopes followed by an oxygenation event has been suggested as the first stage in biomineralisation (Miot et al., 2009; Levett et al., 2020a). Although fossilised bacteria have been demonstrated in canga and are hypothesised to cement detrital fragments (Levett et al., 2021), this is the first evidence of live biofilm mineralisation in the active root-rock interface.

The importance of plant-induced micro-environments and fluctuating conditions for canga formation is clearly evident in the co-occurrence of (i) fresh biofilm mineralisation coating iron-depleted rims (Fig. 5B-D), (ii) chemical precipitation on top of weathered fossilised biofilms (Fig. 5E-G) and (iii) dissolution features (not shown), all within the same 2 mm diameter root channel. This repetitive dissolution and reprecipitation in canga's transient zone appears to be caused by a shift in chemical conditions (*i.e.*, cessation in the secretion of root exudates, percolation of oxygenic water, aeration due to drying), primarily driven by roots. The sedges' short-lived perennial life cycles control carbon supply and moisture retention in the rhizosphere. Microscale anaerobic environments within overall oxic conditions are likely to be critical for iron biogeochemical cycling in canga (Gagen et al., 2018; Levett et al., 2020b). The outcome of the net-generation of iron oxides in the system is a surface duricrust extremely resistant to weathering. Engineered replications of the natural iron duricrust formation processes has been trialed at laboratory- and pilot-scale as a stabilisation strategy for post-mining restoration (Gagen et al., 2020; Paz et al., 2021), which provide a strategy for the bioremediation or iron ore mines in montane regions (Levett et al., 2022). We suggest that plants with strong exudation pattern could facilitate mentioned restoration attempts by supplying carbon to the system and enhancing seasonal biogeochemical cycles and niche heterogeneity.

5. Conclusion

The geochemical, molecular, and microscopic evidence presented here highlights the involvement of plants from the sedge family in the present-day cycling of iron and aluminium in canga outcrops in Carajás, Brazil. Root channels act as hotspots for the opposing processes of mineral dissolution and reprecipitation, concurring in different microenvironments of the rhizosphere. Goethite and hematite dissolution by plant exudates is likely driven by the plant need for nutrients such as phosphorus, resulting in excess soluble iron in the pore water solution. The surplus iron not taken up by the plants or the rhizosphere microorganisms can reprecipitate as iron oxides, coating the walls of inactive root channels, fossilising biofilms, and cementing exogenous fragments. Plants also promote aluminium enrichment at the root-rock interface by (i) preferentially dissolving iron, leading to the relative enrichment of aluminium, and (ii) enabling the crystallisation of secondary gibbsite as cutans. The presence of residual iron(II) under overall aerobic conditions and the detection of potential iron-reducing bacteria in the rhizosphere highlights the importance of biotic iron reduction in these microenvironments, enhancing mineral dissolution and iron mobilisation. Oscillating conditions in the rhizosphere, driven by plants, control the current biogeochemical cycling of iron and contribute to the complex pedogenic nature of canga. The potential contribution of the plant-microorganism-rock interactions in iron- and aluminium-rich duricrusts should not be overlooked with respect to future remediation attempts of iron ore mined areas.

CRediT authorship contribution statement

Anat Paz: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Emma J. Gagen:** Investigation, Resources, Software, Supervision, Writing – review & editing. **Alan Levett:** Conceptualization, Formal analysis, Investigation, Writing – review & editing. **Michael W.M. Jones:** Methodology, Software. **Peter M. Kopittke:** Conceptualization, Supervision, Writing – review & editing. **Gordon Southam:** Conceptualization, Funding acquisition, Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.scitotenv.2024.170119.

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